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-: Assignment 3: -

SQL Queries- All types of Joins, Sub-Query and View:

```
mysql> CREATE DATABASE Employee_Appraisal_System;
Query OK, 1 row affected (0.14 sec)

mysql> USE Employee_Appraisal_System;
Database changed
mysql> CREATE TABLE Departments (
    ->     dept_id INT PRIMARY KEY,
    ->     dept_name VARCHAR(50) NOT NULL
    -> );
Query OK, 0 rows affected (0.16 sec)

mysql> CREATE TABLE Employees (
    ->     emp_id INT PRIMARY KEY,
    ->     name VARCHAR(50) NOT NULL,
    ->     dept_id INT,
    ->     designation VARCHAR(50),
    ->     salary DECIMAL(10,2),
    ->     FOREIGN KEY (dept_id) REFERENCES Departments(dept_id)
    -> );
Query OK, 0 rows affected (0.13 sec)

mysql> CREATE TABLE Managers (
    ->     manager_id INT PRIMARY KEY,
    ->     manager_name VARCHAR(50),
    ->     dept_id INT,
    ->     FOREIGN KEY (dept_id) REFERENCES Departments(dept_id)
    -> );
Query OK, 0 rows affected (0.13 sec)

mysql> CREATE TABLE Appraisals (
    ->     appraisal_id INT PRIMARY KEY,
    ->     emp_id INT,
    ->     review_date DATE,
    ->     rating INT CHECK (rating BETWEEN 1 AND 5),
    ->     bonus DECIMAL(10,2),
    ->     FOREIGN KEY (emp_id) REFERENCES Employees(emp_id)
    -> );
Query OK, 0 rows affected (0.13 sec)
```

```
mysql> INSERT INTO Departments VALUES
    -> (1, 'HR'),
    -> (2, 'IT'),
    -> (3, 'Finance'),
    -> (4, 'Marketing');
Query OK, 4 rows affected (0.06 sec)
Records: 4  Duplicates: 0  Warnings: 0

mysql> INSERT INTO Employees VALUES
    -> (101, 'Amit', 2, 'Developer', 60000),
    -> (102, 'Sneha', 1, 'HR Manager', 50000),
    -> (103, 'Rahul', 2, 'Tester', 45000),
    -> (104, 'Priya', 3, 'Accountant', 55000),
    -> (105, 'Karan', 4, 'Marketing Exec', 40000);
Query OK, 5 rows affected (0.05 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql> INSERT INTO Managers VALUES
    -> (201, 'Mr. Sharma', 2),
    -> (202, 'Ms. Iyer', 1),
    -> (203, 'Mr. Mehta', 3),
    -> (204, 'Ms. Kapoor', 4);
Query OK, 4 rows affected (0.01 sec)
Records: 4  Duplicates: 0  Warnings: 0

mysql> INSERT INTO Appraisals VALUES
    -> (1, 101, '2024-01-15', 5, 10000),
    -> (2, 102, '2024-01-20', 4, 8000),
    -> (3, 103, '2024-01-18', 3, 5000),
    -> (4, 104, '2024-01-25', 4, 7000),
    -> (5, 101, '2025-01-10', 5, 12000);
Query OK, 5 rows affected (0.01 sec)
Records: 5  Duplicates: 0  Warnings: 0
```

```
mysql> SELECT * FROM Employees;
+--------+-------+---------+----------------+----------+
| emp_id | name  | dept_id | designation    | salary   |
+--------+-------+---------+----------------+----------+
|    101 | Amit  |       2 | Developer      | 60000.00 |
|    102 | Sneha |       1 | HR Manager     | 50000.00 |
|    103 | Rahul |       2 | Tester         | 45000.00 |
|    104 | Priya |       3 | Accountant     | 55000.00 |
|    105 | Karan |       4 | Marketing Exec | 40000.00 |
+--------+-------+---------+----------------+----------+
5 rows in set (0.05 sec)

mysql> SELECT * FROM Departments;
+---------+-----------+
| dept_id | dept_name |
+---------+-----------+
|       1 | HR        |
|       2 | IT        |
|       3 | Finance   |
|       4 | Marketing |
+---------+-----------+
4 rows in set (0.00 sec)

mysql> SELECT * FROM Appraisals;
+--------------+--------+-------------+--------+----------+
| appraisal_id | emp_id | review_date | rating | bonus    |
+--------------+--------+-------------+--------+----------+
|            1 |    101 | 2024-01-15  |      5 | 10000.00 |
|            2 |    102 | 2024-01-20  |      4 |  8000.00 |
|            3 |    103 | 2024-01-18  |      3 |  5000.00 |
|            4 |    104 | 2024-01-25  |      4 |  7000.00 |
|            5 |    101 | 2025-01-10  |      5 | 12000.00 |
+--------------+--------+-------------+--------+----------+
5 rows in set (0.00 sec)

mysql> SELECT * FROM Managers;
+------------+--------------+---------+
| manager_id | manager_name | dept_id |
+------------+--------------+---------+
|        201 | Mr. Sharma   |       2 |
|        202 | Ms. Iyer     |       1 |
|        203 | Mr. Mehta    |       3 |
|        204 | Ms. Kapoor   |       4 |
+------------+--------------+---------+
```

a. **Write at least 10 SQL queries for suitable database application using SQL DML.**

- **INNER JOIN – Employees with their department names**

```
mysql> SELECT e.name, d.dept_name
    -> FROM Employees e
    -> INNER JOIN Departments d
    -> ON e.dept_id = d.dept_id;
+-------+-----------+
| name  | dept_name |
+-------+-----------+
| Sneha | HR        |
| Amit  | IT        |
| Rahul | IT        |
| Priya | Finance   |
| Karan | Marketing |
+-------+-----------+
5 rows in set (0.00 sec)
```

- **LEFT JOIN – All employees (including those without departments)**

```
mysql> SELECT e.name, d.dept_name
    -> FROM Employees e
    -> LEFT JOIN Departments d
    -> ON e.dept_id = d.dept_id;
+-------+-----------+
| name  | dept_name |
+-------+-----------+
| Amit  | IT        |
| Sneha | HR        |
| Rahul | IT        |
| Priya | Finance   |
| Karan | Marketing |
+-------+-----------+
5 rows in set (0.00 sec)
```

- **RIGHT JOIN – All departments with employees**

```
mysql> SELECT e.name, d.dept_name
    -> FROM Employees e
    -> RIGHT JOIN Departments d
    -> ON e.dept_id = d.dept_id;
+-------+-----------+
| name  | dept_name |
+-------+-----------+
| Sneha | HR        |
| Amit  | IT        |
| Rahul | IT        |
| Priya | Finance   |
| Karan | Marketing |
+-------+-----------+
5 rows in set (0.00 sec)
```

- **Subquery – Employees earning above average salary**

```
mysql> SELECT name, salary
    -> FROM Employees
    -> WHERE salary > (SELECT AVG(salary) FROM Employees);
+-------+----------+
| name  | salary   |
+-------+----------+
| Amit  | 60000.00 |
| Priya | 55000.00 |
+-------+----------+
2 rows in set (0.05 sec)
```

- **Correlated Subquery – Employees earning more than department average**

```
mysql> SELECT e.name, e.salary
    -> FROM Employees e
    -> WHERE e.salary > (
    ->     SELECT AVG(salary)
    ->     FROM Employees
    ->     WHERE dept_id = e.dept_id
    -> );
+------+----------+
| name | salary   |
+------+----------+
| Amit | 60000.00 |
+------+----------+
1 row in set (0.00 sec)
```

- **JOIN + Subquery – Highest paid employee in each department**

```
mysql> SELECT e.name, e.salary, e.dept_id
    -> FROM Employees e
    -> WHERE e.salary = (
    ->     SELECT MAX(salary)
    ->     FROM Employees
    ->     WHERE dept_id = e.dept_id
    -> );
+-------+----------+---------+
| name  | salary   | dept_id |
+-------+----------+---------+
| Amit  | 60000.00 |       2 |
| Sneha | 50000.00 |       1 |
| Priya | 55000.00 |       3 |
| Karan | 40000.00 |       4 |
+-------+----------+---------+
4 rows in set (0.00 sec)
```

b. Write queries using group by and having clause.

- **Total salary per department**

```
mysql> SELECT dept_id, SUM(salary) AS total_salary
    -> FROM Employees
    -> GROUP BY dept_id;
+---------+--------------+
| dept_id | total_salary |
+---------+--------------+
|       1 |     50000.00 |
|       2 |    105000.00 |
|       3 |     55000.00 |
|       4 |     40000.00 |
+---------+--------------+
4 rows in set (0.00 sec)
```

- **Average salary per department**

```
mysql> SELECT dept_id, AVG(salary) AS avg_salary
    -> FROM Employees
    -> GROUP BY dept_id;
+---------+--------------+
| dept_id | avg_salary   |
+---------+--------------+
|       1 | 50000.000000 |
|       2 | 52500.000000 |
|       3 | 55000.000000 |
|       4 | 40000.000000 |
+---------+--------------+
4 rows in set (0.00 sec)
```

- **Departments with more than 3 employees**

```
mysql> SELECT dept_id, COUNT(*) AS emp_count
    -> FROM Employees
    -> GROUP BY dept_id
    -> HAVING COUNT(*) > 3;
Empty set (0.05 sec)
```

- **Departments with average salary greater than 50,000**

```
mysql> SELECT dept_id, AVG(salary) AS avg_salary
    -> FROM Employees
    -> GROUP BY dept_id
    -> HAVING AVG(salary) > 50000;
+---------+--------------+
| dept_id | avg_salary   |
+---------+--------------+
|       2 | 52500.000000 |
|       3 | 55000.000000 |
+---------+--------------+
2 rows in set (0.00 sec)
```

\

- **Total bonus per department**

```
mysql> SELECT e.dept_id, SUM(a.bonus) AS total_bonus
    -> FROM Employees e
    -> JOIN Appraisals a ON e.emp_id = a.emp_id
    -> GROUP BY e.dept_id;
+---------+-------------+
| dept_id | total_bonus |
+---------+-------------+
|       1 |     8000.00 |
|       2 |    27000.00 |
|       3 |     7000.00 |
+---------+-------------+
3 rows in set (0.00 sec)
```

- **Departments with average rating > 4**

```
mysql> SELECT e.dept_id, SUM(a.bonus) AS total_bonus
    -> FROM Employees e
    -> JOIN Appraisals a ON e.emp_id = a.emp_id
    -> GROUP BY e.dept_id;
+---------+-------------+
| dept_id | total_bonus |
+---------+-------------+
|       1 |     8000.00 |
|       2 |    27000.00 |
|       3 |     7000.00 |
+---------+-------------+
3 rows in set (0.00 sec)
```